UNIVERSITY OF BAGHDAD

COLLEGE OF MEDICINE

2026-2027

LECTURE TITLE

Inorganic Nutrients; Minerals

Department of Family and Community Medicine

Third Stage 2026–2027

Lecturer Dr. Maysaa Adel Kadhim

Learning Objectives

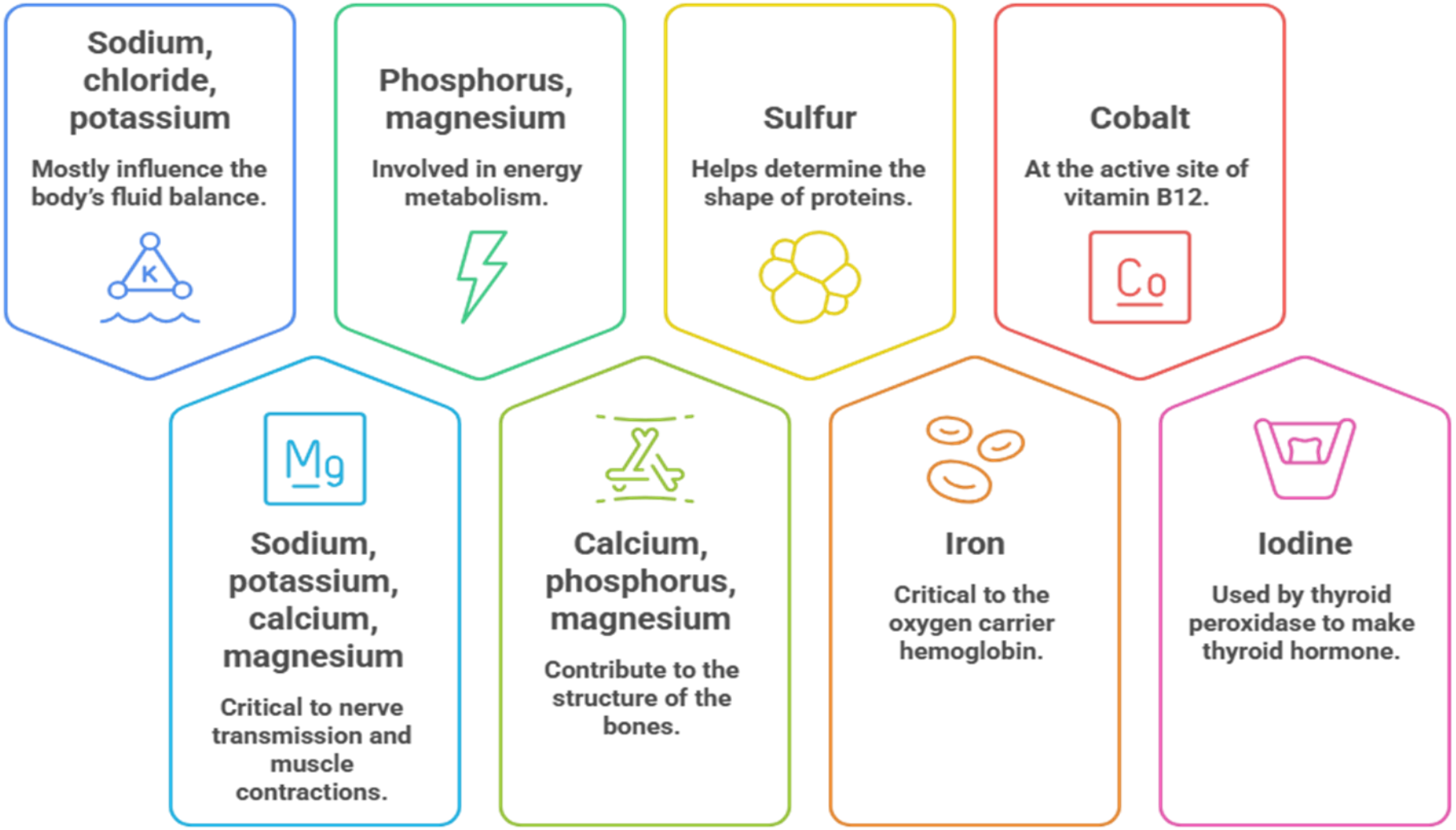
Clear, measurable objectives with a premium academic hierarchy.

- Define and classify major and trace minerals.
- Describe the functions and dietary sources of iron and iodine.
- Identify the risk groups, and manifestations of iron and iodine deficiencies.
- Explain the main approaches to the diagnosis, prevention, and control of iron and iodine deficiencies.

- **Major minerals:** essential mineral nutrients required in the adult diet in amounts greater than 100 milligrams per day.
- **Trace minerals:** essential mineral nutrients required in the adult diet in amounts less than 100 milligrams per day.
- Because the body cannot make minerals, all minerals must be consumed in the foods that we eat.

MAJOR MINERALS	TRACE MINERALS
Calcium	Chromium
Chloride	Copper
Magnesium	Fluoride
Phosphorus	Iodine
Potassium	Iron
Sodium	Manganese
Sulfur	Molybdenum
	Selenium
	Zinc

Functions of the minerals in the body



Hidden Hunger



VITAMIN A



IRON



IODINE





ENOUGH FOOD BUT STILL MISSING NUTRIENTS

Iron

- Iron is a nutrient found in trace amounts in every cell of the body.
- Iron is essential for life, but it can be toxic in excess.
- Iron is part of hemoglobin in red blood cells and myoglobin in muscles. The role of both of these molecules is to carry oxygen. Iron also makes up part of many proteins and enzymes in the body.
- Iron is transported in the body bound to transferrin, and it is stored as ferritin in the liver, the spleen, and other tissues.



Iron deficiency anemia (IDA)

- Anemia is a condition in which the number of red blood cells or the hemoglobin concentration within them is lower than normal.
- Iron-deficiency anemia is the most prevalent nutrition problem in the world today.
- It is a serious global public health problem that particularly affects **children under 5 years of age, particularly infants and children under 2 years of age, menstruating adolescent girls and women, and pregnant and postpartum women.**
- **Low- and lower-middle income countries** bear the greatest burden of anemia, particularly affecting **populations living in rural settings, in poorer households and who have received no formal education.**

How do iron deficiency and iron-deficiency anemia differ conceptually?

Iron deficiency involves depleted iron stores, whereas iron-deficiency anemia is a late manifestation marked by low hemoglobin concentration.

WHO classification of anemia according to age and severity

Age groups	Hb level (gm/dl)			
	No anemia	Mild	Moderate	Severe
Children 6-59 months of age	≥ 11	10-10.9	7-9.9	< 7
Children 5-11 years of age	≥ 11.5	11-11.4	8-10.9	< 8
Children 12-14 years of age	≥ 12	11-11.9	8-10.9	< 8
Non pregnant women(≥15years of age)	≥ 12	11-11.9	8-10.9	< 8
Pregnant women	≥ 11	10-10.9	7-9.9	< 7
Men	≥ 13	11-12.9	8-10.9	< 8

Cut-off values for public health significance

Indicator	Prevalence cut-off values for public health significance estimated from blood levels of hemoglobin or hematocrit	
Anemia	< 5%	No public health problem
	5-19%	Mild public health problem
	20-39%	Moderate public health problem
	≥ 40%	Severe public health problem

What are the consequences and implications?

Poor pregnancy outcome (increase maternal, fetal mortality, premature birth, low birth weight, and postpartum health issues).

Poor school performance (developmental delays and behavioural disturbances such as decreased motor activity, social interaction and attention to tasks)

Reduced work capacity of individuals and entire populations, with serious consequences for the economy and national development.

Impaired immunity (high risk of infection).

Associated risk of lead poisoning because of pica.

Causes of iron deficiency anemia

Inadequate Ingestion

Poor diet without supplementation of iron

Inadequate Absorption

Diarrhea, achlorhydria, intestinal disease such as celiac disease, atrophic gastritis, partial or total gastrectomy, or drug interference

Inadequate Utilization

Chronic gastrointestinal disturbances

Increased Requirement

Increase of blood volume, which occurs during infancy, adolescence, pregnancy, and lactation and which is not being matched with intake

Increased Excretion

Excessive menstrual blood, hemorrhage from injury, or chronic blood loss from various medical conditions

Increased Destruction

Caused by a chronic inflammation or other chronic disorder leading to destruction of RBC's

Clinical features of iron deficiency anemia

1. General

- Fatigue & weakness
- Reduced exercise tolerance
- Headache / dizziness
- Poor appetite
- Irritability

2. Skin & Appendages

- Pallor (conjunctiva, nail beds, palms & soles)
- Cold extremities (hands and feet)
- Hair loss
- Brittle nails , koilonychia (spoon-shaped nails)



Clinical features of iron deficiency anemia

3. Oral & Gastrointestinal

- Glossitis (smooth, sore, red tongue)



- Angular stomatitis

(fissuring and ulceration at the angles of the mouth)



- Cheilosis (swelling and fissuring of the lips)



- Dysphagia

Clinical features of iron deficiency anemia

4. Neurological / Behavioral

- **PICA/ pagophagia:** An unusual craving for non edible substances such as:

  -Ice -Dirt -Paint or starch.

- Restless legs syndrome
- Cognitive impairment, particularly in children

5. Cardiopulmonary

- Dyspnea, especially on exertion and rest
- Palpitations / tachycardia
- Systolic flow murmur
- Postural hypotension
- Severe cases → heart failure

Laboratory diagnosis

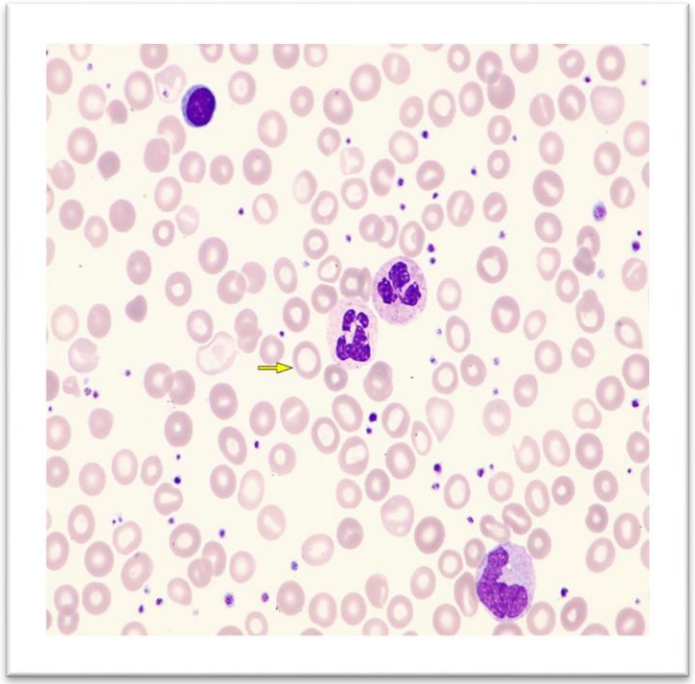
- **Complete blood count:**

- Low Hemoglobin(Hb) level
- Hypochromic microcytic anemia.

- **Iron profile:**

- Low S. iron - Low S. ferritin
- High Total Iron-Binding Capacity (TIBC)& transferrin

S. ferritin "**Most sensitive tool** for evaluation of iron status, especially in populations with low prevalence of anemia".



Source of iron in food

Iron occurs in two forms in foods: heme and nonheme.

Heme iron:

- Is more bioavailable and more efficiently absorbed than nonheme iron **(15 to 35 %)**.
- **Not influenced** by most inhibitory factors in the diet.
- Is found in **only 40% of the animal food sources** and in **no plant foods**.
- Animal sources of heme iron are meat, poultry, fish, seafood, organ meats and eggs (especially the yolks).



Nonheme iron:

- Is less efficiently absorbed than heme iron, because it is more tightly bound in foods **(2-20%).**
- Is found in **60% of the animal food sources** and **all the plant food sources**.
- Plant Sources of nonheme iron are iron-fortified cereals, dried beans, whole grains, legumes, nuts, dried fruits, vegetables, and green leafy vegetables.
- Animal sources of nonheme iron are eggs, dairy products, poultry.



Inhibitors of iron absorption

Phytates (Phytic Acid): Found in whole grains, beans, lentils, nuts, and seeds; binds strongly to iron to block uptake.

Polyphenols and Tannins: Found in black and herbal tea, coffee, cocoa, whole grain, cereals, and some vegetables; form insoluble complexes with iron.

Antacids / Gastric Acid Inhibitors: Raising stomach pH reduces the conversion of iron into its most absorbable ferrous form.

Calcium: Found in dairy products and supplements; interferes with both non-heme and heme iron absorption via competitive pathways.

Pharmacological dose and supplements of Zinc.

Oxalic acid or oxalate: found in spinach, chard, beans, nuts and chocolate.

Enhancer of iron absorption

Vitamin C (Ascorbic Acid): A powerful enhancer for nonheme iron and should be consumed at the same meal . It binds with ferric iron to form a soluble chelate that remains stable in the alkaline environment of the duodenum. It also reduces ferric iron (Fe^{3+}) to the more easily absorbed ferrous form (Fe^{2+}).

Some Animal Proteins: meat, fish and poultry proteins **(MFP Factor)** stimulates nonheme iron absorption.

Recommended Dietary Allowances for Iron for Children and Adults

Age	Male	Female	Pregnancy	Lactation
0-6 months	0.27 mg*	0.27 mg*		
7-12 months	11 mg	11 mg		
1-3 years	7 mg	7 mg		
4-8 years	10 mg	10 mg		
9-13 years	8 mg	8 mg		
14-18 years	11 mg	15 mg	27 mg	10 mg
19-50 years	8 mg	18 mg	27 mg	9 mg
51+ years	8 mg	8 mg		

*Adequate intake (AI)

Tolerable upper intake level (UL): 0-12 mos 40 mg; 1-3y 40 mg/d; 4-8y 119 mg/d; 9-13y 40 mg/d; 14-18y (also for pregnancy and lactation) 45 mg/d; 19+ (also for pregnancy and lactation) 45 mg/d.

Prevention & Control of Iron Deficiency Anaemia (IDA)

Better nutrition → Stronger health → Brighter future

1 Dietary Improvement

Iron-rich + Vitamin C



- Eat iron-rich foods (red meat, fish, poultry, legumes, fortified cereals, dark green vegetables).
- Combine with vitamin C-rich foods (fruits & vegetables) to enhance absorption.
- Reduce inhibitors (tea, coffee, cocoa, calcium) around meals when possible.

2 Food Fortification

Iron-fortified staple foods



- Examples: flour, cereals and other staple foods.

3 Iron Supplementation

High-risk groups



- Provide iron supplements (e.g. ferofolic) for pregnant and lactating women and other high-risk groups.

4 Other Complementary Public Health Measures

- Promote breastfeeding & use of iron-rich complementary foods.
- Reduce infectious diseases & hookworm control.
- Immunization.
- Environmental health.
- Control of micronutrient malnutrition.
- Community-based primary health care (antenatal care, nutrition centers, neonatal care, IMCI, family planning...).

5 Screening, Monitoring & Evaluation



- Monitor anemia and iron status in target populations.
- Assess the effectiveness of supplementation and fortification programs.

Key markers:

- Hemoglobin
- Ferritin (iron stores)
- Other relevant indicators

Iodine

- It is an important mineral that is naturally found in some foods and added to others (**primarily iodized salt**).
- It is an essential component of thyroid hormones, which plays an important role in physical and mental development.
- Iodine deficiency is a major cause of **preventable mental retardation and brain damage in the world.**



Iodine deficiency disorders (IDD)

IDD refer to a spectrum of health consequences resulting from inadequate intake of iodine, and they are:

- **Goiter**
- **Congenital hypothyroidism** is a congenital disorder resulting from insufficient thyroid hormone to the fetus during gestation.
- **Cretinism** is the severe, irreversible syndrome of stunted physical and mental growth due to untreated congenital hypothyroidism.







- **Impaired mental and physical development.**

- **Hypothyroidism** occurs when a poorly functioning thyroid gland does not make enough T4, thereby greatly reducing the basal metabolic rate.

- **Increased morbidity and mortality of infants and neonates**.

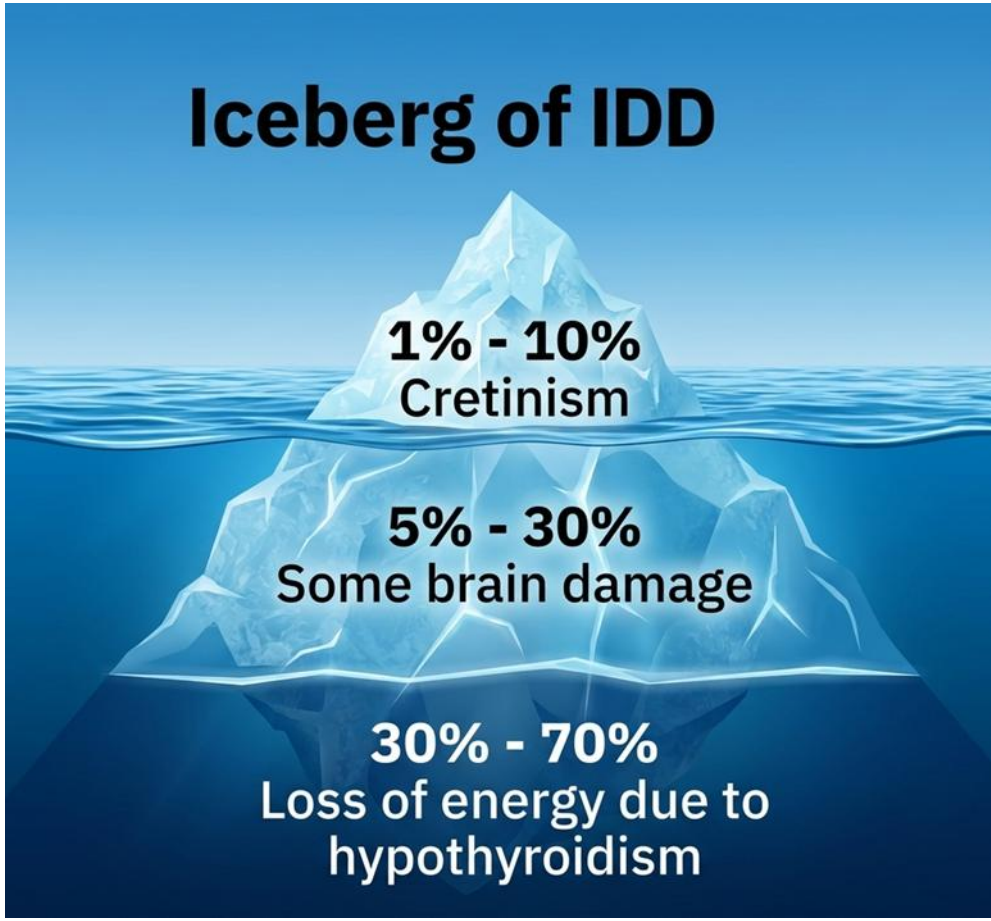


Symptoms & Signs of Hypothyroidism

Epidemiology

Iodine deficiency is a major public health problem for populations throughout the world, particularly for pregnant women and young children.

According to WHO a goiter prevalence of 5% or more in school age children is an indication of iodine deficiency in a population.



Indicators of IDD assessment

Indicators	Goal
Enlarged thyroid in school children 6-12 years of age by palpation or ultrasound	< 5%
Proportion of households consuming adequately iodized salt	> 90%
Neonatal TSH proportion with levels > 5 ml U/L whole blood obtained within 72 hours from birth	< 3%

Groups at risk of iodine deficiency

- Pregnant and lactating women
- Fetuses, newborns, and preschool children
- People who do not use iodized salt
- People with inadequate dietary iodine intake
- Vegans and people with low intake of dairy products, seafood, and eggs
- People with marginal iodine status and high intake of goitrogenic foods (e.g., cabbage, broccoli, cauliflower), which may interfere with iodine uptake
- People with selenium deficiency – may impair thyroid hormone metabolism and increase vulnerability to iodine deficiency

Recommended Dietary Allowances (RDAs) for Iodine

Age	Male mcg/day	Female mcg/day	Pregnancy mcg/day	Lactation mcg/day
0-6 months	110*	110*	N/A	N/A
7-12 months	130*	130*	N/A	N/A
1-3 years	90	90	N/A	N/A
4-8 years	90	90	N/A	N/A
9-13 years	120	120	N/A	N/A
14-18 years	150	150	220	290
19+	150	150	220	290

N/A, Not applicable.

*Adequate intake (AI).

Tolerable upper intake level (UL): 0-12 mos not determined; 1-3y 200 mcg/d; 4-8y 300 mcg/d; 9-13y 600 mcg/d; 14-18y (also for pregnancy and lactation) 900 mcg/d; 19+ (also for pregnancy and lactation) 1, 100 mcg/d.

Iodine Deficiency Disorders (IDD)
Prevention and Control

Healthy iodine status... for a healthier future

1 Fortify
(Universal Salt Iodization)



- Use iodized salt for household and food processing.
- Indine concentration at production: 20–40 mg/kg (ppm) (adjust for salt intake and iodine losses).
- Also fortify other foods (flour, milk, water, condiments) when needed.

2 Supplement
(Targeted)



- For high-risk groups:
  - Pregnant & lactating women
  - Infants and young children
  - Other vulnerable groups (when dietary intake is insufficient).
- Via capsules or iodized oil (as needed).

3 Educate
(Health Promotion)



- Raise awareness about the consequences of IDD, especially for high-risk groups (infants, pregnant and lactating mothers).
- Advise the use of Iodized salt.
- Encourage iodine-rich foods (sea fish, shelffish, shrimp, dairy, eggs).
- Avoid goitrogenic foods.

4 Detect
(Early Diagnosis)



Diagnostic tests to detect IDD:

- Serum total and free T_3, free T_4, and sensitive TSH levels;
- Urinary iodine concentration (UIC);
- Heel prick test in neonates for congenital hypothyroidism.

5 Monitor
(Surveillance)



- Monitor the distribution of adequately iodized salt in the community.
- Regularly assess population iodine status (e.g., UIC).

6 Support
(Political)



- Provide resources for iodization programs.
- Raise political attention to the problem of IDD (loss of productivity & quality of life).

Prevent → Protect → Educate → Detect → Monitor → Support

Source: WHO. Prevention and control of iodine deficiency in the WHO European Region, 2024.

Summary

- Minerals are essential micronutrients required for normal physiological functions.
- Iron is essential for oxygen transport; its deficiency causes iron-deficiency anemia.
- Iodine is essential for thyroid hormone synthesis, growth, and development.
- Adequate dietary intake, fortification, supplementation, and monitoring are key to preventing mineral deficiencies.

References

Keep the source list consistent and academically formatted.

1. DeBruyne L et al. Nutrition & Diet therapy. Cengage.2020

2. Staci N, et al. Williams' basic nutrition and diet therapy. Elsevier.2017

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QUESTIONS & DISCUSSION

Thank you

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LECTURE COMPLETE